Explainable AI for ESG-Driven Financial Risk Analysis

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***Abstract*— In recent times, environmental, social, and governance (ESG) metrics are being used in financial decision making, but the structural relationship with types of financial risk remains poorly understood. The proposed learning framework helps to assess how ESG factors have a bearing on two financial risk dimensions i.e. short-term market volatility risk and long-term return risk. Daily stock price data of S&P 500 companies and ESG scores are used to construct binary risk labels for volatility and cumulative returns. Gradient boosting classifiers (LightGBM) are used to model these risk targets and use Shapley Additive Explanations (SHAP) to explain them. Our findings suggest that ESG indicators play a weak role in explaining short-term volatility, but they have a somewhat stronger structural role in long-return risk. The breakdown of ESG dimensions suggests differing levels of contribution for the Environmental, Social and Governance components. Additionally, the interaction analysis shows sector-specific contribution for ESG effect. The proposed method offers a transparent approach to analysing ESG-risk. It comprehensively insights into how the sustainability-related factor informs financial risk assessors. Ultimately, it has profound implications for investors, policymakers and corporate risk-takers.**

I. INTRODUCTION

Environmental, social, and governance factors have taken on increasing importance in today's financial decision-making. As companies become more aware of their impact on the environment and are moving towards a more sustainable approach , they have really begun to factor in indicators such as ESG scores, they consider sustainability-related indicators together with conventional financial indicators while assessing firm's performance and risk. As market participants increasingly regard sustainability as a risk factor and a potential source of market excellence, demand for these investment strategies is clearly growing.

At present, thinking about the financial relevance of ESG is conceptually unclear. Typical financial risk models are often limited solely to market-driven dynamics such as price volatility and liquidity, and not systemic macroeconomic conditions. Typical ESG indicators, meanwhile, represent slower-moving structural characteristics of firms which tend to affect them in long term. This brings up the the question , how do ESG factors relate to different financial risk and are they necessarily related to all forms of risk?

Recent advances in machine learning have helped us incorporate these factors in our risk models however, many of these existing approaches rely on black-box predictive systems because of which they have limited interpretability, making it difficult to co-relate how specific ESG factors influence model's outcomes. Moreover, most studies treat ESG as a single aggregated score, which obscures the distinct roles of Environmental, Social, and Governance factors and reduces analytical clarity.

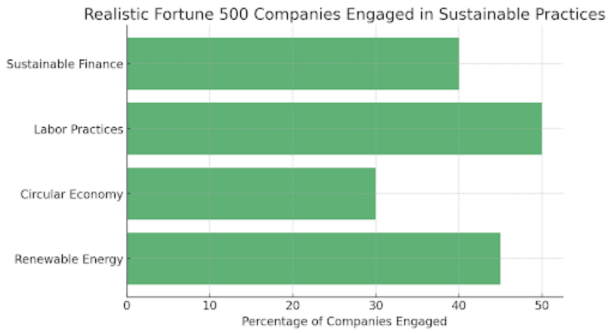


Fig. 1: Realistic Fortune 500 Companies Engaged in Sustainable Practices

The machine learning approach that we used was unique as it took in the ESG score not as a singular entity but as individual factors and studied their individual impact on a particular stock. We considered two types of risks that have different structures: short-term volatility risk and long-term return risk. By exploiting the stock price data of S&P 500 firms and the ESG indicators, we introduce gradient boosting models and SHAP for transparent and interpretable feature level risk formation. Dividing ESG into its parts and studying how those interact is a starting point for understanding beyond prediction. Using the sector-reduced forms of our model, we get episodes of interpretable insight in terms of structural analysis. It discloses how ESG interacts with firms' financial risk in various market environments.

II. LITERATURE REVIEW

A broad and growing body of research in data science, sustainability studies, and finance has resulted from the increasing incorporation of ESG considerations into financial analysis. Previous research has covered explainability frameworks, machine learning applications, empirical financial analysis, and theoretical underpinnings. With an emphasis on ESG and financial performance, ESG and risk modeling, machine learning in financial risk assessment, and the rise of explainable artificial intelligence in finance, this section examines the major lines of literature pertinent to this investigation. The main research gaps that drive the current study are identified in the section's conclusion.

A. Financial Performance and ESG

Stakeholder theory and corporate social responsibility frameworks gave rise to early ESG research, which mainly looked at sustainability practices in relation to firm value, reputation, and ethical investment. Some early empirical research found weak or negligible relationships between ESG performance and financial returns, while other studies found positive associations. Although these effects are still very context-dependent across sectors, markets, and time horizons, more recent meta-analyses and extensive empirical studies indicate that companies with stronger ESG profiles typically show better long-term performance, increased resilience, and reduced exposure to downside risk. Although this corpus of work identifies ESG as a potentially significant factor in financial outcomes, it lacks a coherent and consistent explanatory framework.

B. ESG and Financial Risk Modeling

Later research looked at the connection between various types of financial risk and ESG indicators, shifting the emphasis from performance to risk. According to research, companies with weaker governance or social practices are more susceptible to extreme negative events. ESG scores have also been linked to downside risk, crash risk, and tail-risk exposure. While some studies stress environmental resilience in long-term risk mitigation, others highlight the role of governance in lowering short-term instability. Additionally, sectoral heterogeneity has been noted, with ESG impacts varying significantly among sectors like manufacturing, technology, finance, and energy. Many of these studies, however, treat financial risk as a single construct without making a clear distinction between structurally distinct risk types, such as long-term performance risk and short-term volatility.

C. Machine Learning in Financial Risk Assessment

High-dimensional financial and non-financial data can now be integrated thanks to the quick growth of machine learning's use in financial risk modeling. In tasks like credit risk classification, return prediction, and volatility forecasting, models like gradient boosting, random forests, and deep learning architectures have shown enhanced predictive performance. Financial variables and ESG indicators are combined in hybrid models, which have demonstrated promise in identifying nonlinear relationships that conventional econometric models are unable to identify. However, interpretability, transparency, and a structural understanding of model behavior are not given as much weight as predictive accuracy when evaluating these methods.

D. Explainable Artificial Intelligence in Finance

The opacity of black-box models in high-stakes industries like finance has led to the development of explainable AI (XAI) techniques. By assigning model predictions to specific features, methods like SHAP and LIME allow for both local and global interpretability. Explainability tools have been used in ESG-related research to pinpoint important elements in corporate risk assessments, investment models, and credit scoring. Despite these developments, explainability is applied superficially in the majority of current work, which prioritizes feature ranking over structural interpretation. ESG dimension decomposition, multi-risk modeling, and the analysis of the interactions between ESG factors and sectoral context have received little attention.

E. Research Gaps

There are a number of significant gaps in the literature currently in publication. First, the separate financial roles of the Environmental, Social, and Governance dimensions are often obscured by the fact that ESG is frequently viewed as a composite score. Second, financial risk is frequently modeled as a single construct that does not differentiate between long-term structural risk and short-term market instability. Third, machine learning techniques limit scientific understanding and practical trust by prioritizing predictive performance over interpretability. Lastly, little is known about the sectoral and contextual dependencies of ESG-risk relationships. In order to fill these gaps, this study suggests an interpretable machine learning framework that combines sector interaction analysis, dual-risk modeling, ESG decomposition, and explainable AI techniques to offer a clear and organized understanding of how ESG influences the formation of financial risk.

III. DATA DESCRIPTION AND FEATURE ENGINEERING

In order to create a single analytical dataset, this study combines firm-level ESG indicators with daily stock market data for S&P 500 companies. Trading date, closing price, and firm identifiers are examples of financial data that make it possible to construct return and volatility metrics. In addition to sectoral classifications and controversy indicators, ESG data offer both composite and disaggregated Environmental, Social, and Governance risk scores. To guarantee data consistency and analytical reliability, companies with insufficient ESG information are eliminated and the two datasets are combined using stock ticker symbols.

A. Raw Data Structure

TABLE I: Financial Market Data Fields

Column Name	Description	Type
date	Trading date	Date
close	Daily closing price	Float
ticker	Stock ticker symbol	Categorical

TABLE II: ESG Dataset Fields

Column Name	Description	Type
esg_risk_score	Composite ESG risk score	Float
environment_risk_score	Environmental risk exposure	Float
social_risk_score	Social risk exposure	Float
governance_risk_score	Governance risk exposure	Float
controversy_score	Controversy indicator	Float
sector	Economic sector classification	Categorical
industry	Industry classification	Categorical

B. Data Preprocessing and Feature Engineering

Daily financial returns are computed from stock closing prices using the standard log-return formulation:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t represents the closing price at time t. These daily returns serve as the basis for constructing firm-level financial risk measures.

TABLE III: Financial Feature Engineering

Feature	Formula	Financial Meaning
daily_return	$\ln\left(\frac{P_t}{P_{t-1}}\right)$	Daily price change
volatility	$\mathrm{std}(\text{daily_return})$	Short-term market instability
cumulative_return	$\prod(1 + \text{daily_return}) - 1$	Long-term performance

These aggregated financial indicators form the quantitative foundation for risk labeling and classification tasks in subsequent modeling stages.

C. ESG Feature Representation

Both disaggregated and composite risk indicators are available in the ESG dataset. While the Environmental, Social, and Governance Risk Scores capture risk characteristics specific to each dimension, the composite ESG Risk Score is an overall indicator of a firm's sustainability exposure. Furthermore, metrics pertaining to controversy offer indicators of regulatory and reputational exposure.

This study specifically keeps the disaggregated ESG dimensions as independent features rather than depending only on the composite ESG score. This representation prevents the information loss that comes with score aggregation and permits dimension-level analysis.

TABLE IV: Final Modeling Feature Set

Category	Features
Financial	volatility, cumulative_return
ESG	esg_risk_score, environment_risk_score etc.
Contextual	sector_enc, industry_enc

The final dataset integrates market dynamics, ESG structure, and economic context into a unified feature space suitable for both predictive modeling and explainable analysis. All numerical features are standardized prior to model training to ensure scale consistency and numerical stability.

IV. METHODOLOGY

The model architecture, learning pipeline, and risk construction framework utilized to combine financial market dynamics with ESG indicators are presented in this section. Predictive modeling is in line with explainable AI principles thanks to the methodology's design, which guarantees interpretability, robustness, and structural transparency.

A. Risk Construction Framework

To capture different factors of market risk, two complementary financial risk targets are created:

- Risk of volatility: The standard deviation of daily returns over the course of five years is used to calculate firm-level volatility. In order to guarantee balanced class distributions, companies are then categorized into binary risk categories (Low Risk / High Risk) using percentile-based thresholding.
- Risk Based on Return: The sum of daily returns is used to calculate cumulative returns, which are used to gauge long-term financial performance. Quantile thresholds are also used to classify firms into groups with low and high return risk.

Comparative ESG impact analysis across various financial risk regimes is made possible by the study's dual-risk design, which separates short-term market instability (volatility) from long-term performance risk (returns).

B. Model Architecture

Light Gradient Boosting Machine (LightGBM) classifiers are used to model both risk targets. Because LightGBM can handle heterogeneous feature spaces, model nonlinear feature interactions, and maintain computational efficiency on structured tabular data, it was chosen.

Financial indicators, ESG dimensions, and contextual sectoral variables make up the same unified feature space used by all models. To prevent target leakage and maintain interpretive clarity, separate models are trained for volatility risk and return-based risk.

C. Training and Validation Strategy

To maintain class distributions, stratified sampling is used to divide the dataset into training and testing subsets. To guarantee scale consistency, feature standardization is applied to every numerical variable. A supervised learning framework with cross-entropy loss optimization is used for model training.

To provide a fair assessment of classification quality across risk classes, performance evaluation is carried out using accuracy, precision, recall, F1-score, and confusion matrices.

D. Modeling Objective

This study's methodological goal is to prioritize structural interpretability over pure predictive accuracy. In addition to categorizing financial risk, the models reveal the ways in which sectoral context, financial indicators, and ESG factors interact to influence market risk outcomes. The explainable AI framework created in the next section is directly supported by this design decision.

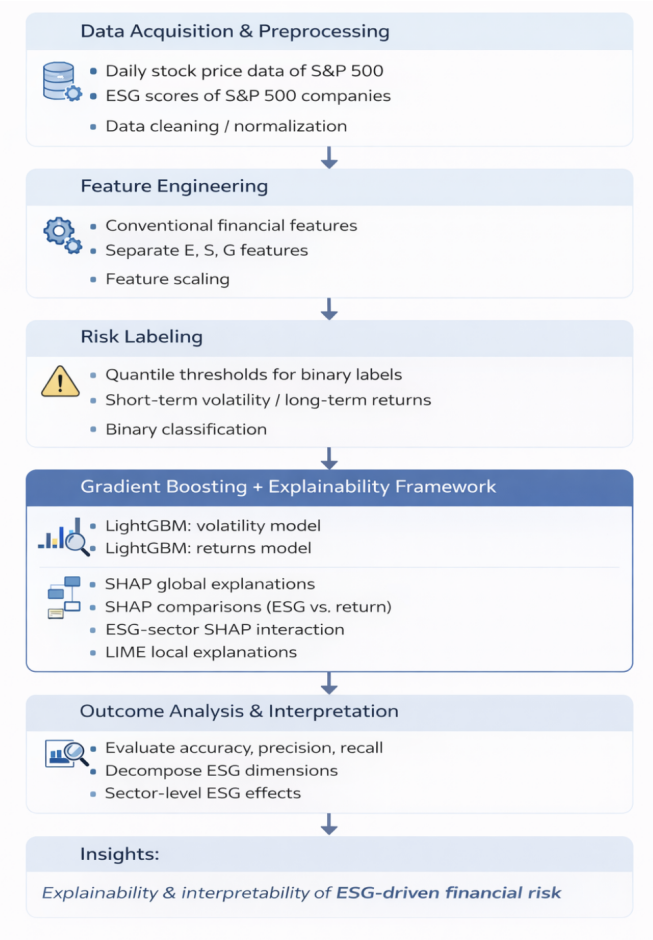


Fig. 2: Methodology for ESG driven risk analysis

V. MODEL PERFORMANCE AND RESULTS

The empirical performance of the suggested risk classification models is shown in this section, along with an assessment of their predictive behavior across tasks involving return-based and volatility-based risk. In addition to evaluating classification accuracy, the goal is to build model reliability as a basis for further explainability analysis.

A. Evaluation Metrics

Accuracy, precision, recall, F1-score, and confusion matrices are among the complementary metrics used to assess the model's performance. This multi-metric evaluation framework prevents inaccurate conclusions based solely on accuracy and guarantees balanced assessment across class distributions.

B. Volatility Risk Model Performance

The classifier for predicting volatility risk provides a relatively moderate accuracy in forecasting due to the highly complex nature of market volatility as well as being reliant upon both financial-driven and ESG-driven influences on volatility.

a) Performance Summary:

- **Accuracy:** 0.58
- **Balanced class performance** across low- and high-volatility categories

TABLE V: Volatility Risk Classification Performance

Metric	Value
Accuracy	0.58
Precision (Low Risk)	0.61
Recall (Low Risk)	0.66
Precision (High Risk)	0.53
Recall (High Risk)	0.48

TABLE VI: Confusion Matrix for Volatility Risk Classifier

	Predicted Low	Predicted High
Actual Low	27	14
Actual High	17	16

b) Confusion Matrix: The findings indicate that volatility risk is, by nature, very difficult to forecast at the firm level based solely on their structural features. This is due to the inherent stochasticity associated with short-term market fluctuations along with the impact of non-firm based macroeconomic factors.

C. Return-Based Risk Model Performance

The return-based risk model show-cases stronger predictive capability in comparision to the volatility model suggesting that ESG and structural firm characteristics have greater impact on the long-term financial performance than on the short-term market instability.

a) Performance Summary:

- **Accuracy:** 0.64
- **Improved class separation** relative to the volatility risk model

TABLE VII: Return Risk Classification Performance

Metric	Value
Accuracy	0.64
Precision (Low Risk)	0.65
Recall (Low Risk)	0.76
Precision (High Risk)	0.62
Recall (High Risk)	0.49

TABLE VIII: Confusion Matrix for Return-Based Risk Classifier

	Predicted Low	Predicted High
Actual Low	39	12
Actual High	21	20

b) Confusion Matrix: These results show that ESG-informed features are more structurally aligned with the long-term performance dynamics than with the short-term volatility patterns.

D. Comparative Performance Analysis

When directly comparing the two models, there are different methods of predicting returns:

- **Volatility risk** has less ability to be predicted because of how much it changes with things like exogenous shocks, macroeconomic activity/events, and the noise created by market microstructure.
- **Return-based risk** has greater ability to be predicted and demonstrates a better relationship between ESG characteristics, firm characteristics, and long-term financial performance.

This difference supports the theory that ESG characteristics act better as long-term risk modifiers vs short-term stabilizers for the overall market.

E. Empirical Interpretation

Instead of treating moderate performance assessments as limits of a model, these results show the inherent complexity associated with all financial risk systems. Financial markets are subject to latent macroeconomic factors, behavioural influences, and geopolitical influences that are not observable in firm-level ESG databases.

As a result, the value of the framework described above is not so much to maximise the degree of predictive accuracy, but rather to help identify the structural relationship between each dimension of ESG (environmental, social and governance) and the establishment of financial risk. This interpretation of results motivates the explainability assessment that follows.

VI. EXPLAINABILITY FRAMEWORK AND INTERPRETATION

This section illustrates the framework used to provide explainability of AI (XAI) utilized for decomposition of risk models and for identifying structural interactions between the various ESG dimensions and how those ESG dimensions form part of financial risk. The goal is to integrate interpretability into the modeling process as an integral part of the analytical objectives rather than providing explainability as an after-the-fact visualization step.

A. Explainability Objective

The explainability framework is designed to support three main objectives:

- **Global Interpretability** Identify what ESG and other contextual features are the strongest drivers of model prediction at the population level.
- **Comparative Interpretability** Assess the respective impacts of ESG dimensions (i.e., types of financial risk) on different types of risk (i.e., volatility vs. return).
- **Local Interpretability** Develop an explanation for the prediction of individual firm-level predictions to provide insight into decision modeling on the basis of an instance of data.

The proposed interpretability model allows authorities to gain both structural and case participation insight.

B. Comparative SHAP Analysis Across Risk Types

Our analysis was based on a comparison of the two risk types that we will analyze—the volatility risk model and the return based risk model. From this comparison we found that the two risk types had significantly different explanatory regimes. In addition, it is worth noting that overall volatility risk is influenced primarily by the social and governance dimensions of ESG, indicating that these factors are sensitive to the reputational stability of an organization, its organizational structure, and short-term levels of institutional trust.

On the other hand, return based risk is significantly affected by the governance and environmental dimensions of ESG, indicating that the level of regulatory exposure to an organization; how well it manages the quality of its strategic governance; and how much it is liable for the future environmental effects of its business activities have a far greater impact on the overall cumulative financial performance of an organization.

Thus, these two risk model comparisons support the hypothesis that the various dimensions of ESG operate on different time scales; i.e., that some ESG factors are related to short-term financial stability of organizations while others directly impact the long-term valuation of organizations.

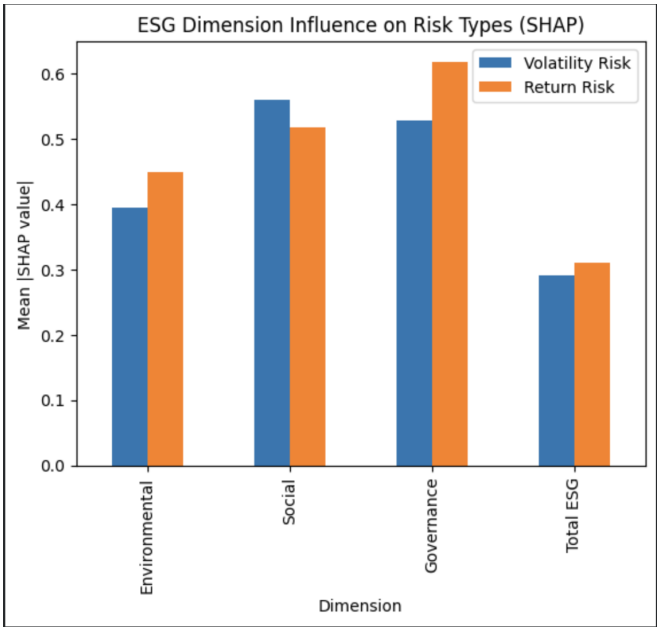


Fig. 3: ESG Dimension impact on Risk types

C. Analysis of ESG Dimensions

The first step to avoiding an interpretive bias in our analysis of combined ESG data would be to decompose the explanatory variables/impacts separately by ESG dimensions (i.e., environmental, social, and governance) using aggregated SHAP value calculations.

In our analysis of the ESG decomposed data, we were clearly able to identify a hierarchical structure in the degree of explanatory variable influence:

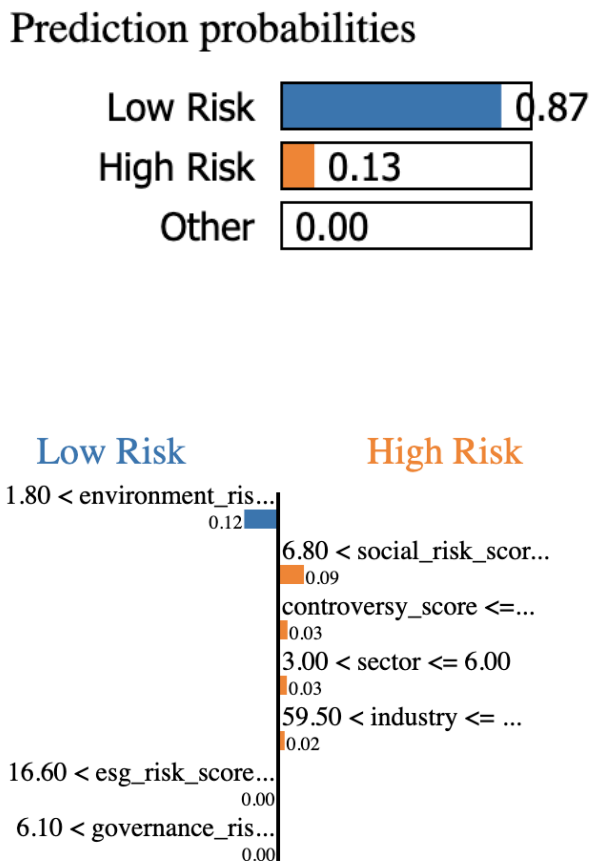
- **Governance risk** is the most consistently significant dimension of risk for both types of risk exposure.
- **Social Risk** The social dimension of ESG has a significant effect on the development of volatility risk.
- **Environmental risk** The environmental dimension of ESG has a significantly greater effect on the development of return-based risk.

In conclusion, the findings of this analysis confirm that the three dimensions of ESG should not be treated as interchangeable or equally significant indicators but rather as a hierarchy of structural importance.

D. ESG × Sector Interaction Analysis

Cross effects were studied using SHAP interaction values for the analysis of ESG dimensions and sectors' contexts. This analysis indicates that the nature of ESG-risk relationships differ widely across sectors.

For example: governance risk has amplified effects in capital-intensive and regulated sectors; environmental risks have substantial explanatory power in resource-dependent sectors; social risks have amplified effects in consumer-facing sectors that are reputationally susceptible to affecting companies' financial stability.



These interactions highlight how ESG factors are not independent of one another; rather, they are rooted in the structures of the sectors and industries where they exist and, as such, affect finances based on their degree of connection to each other.

E. Interpretive Synthesis

The two methods provided an effective combination of complementary interpretive approaches through the use of SHAP for global structural understanding and LIME for localized decision-level insights. Together these methods demonstrate that the ESG dimensions are much more than ancillary predictors and represent significant structural components of financial risk development.

Rather than being considered just a proxy for sustainability branding, the explainability analysis presents ESG as latent risk structures that are embedded in the market dynamics. This transforms ESG from a reporting metric of performance to a risk signal used to link aspects of corporate performance, governance structures, and social stability to the financial mechanisms that create risk.

VII. DISCUSSION

The research indicates that ESG (environmental, social and governance) factors enhance the comprehension of financial risk via interpretability rather than solely through predictive accuracy. Although overall model performance is average, the explanations produced from the analyses reveal meaningful non-linear associations between ESG dimensions and market risk.

ESG factors, however, differ depending on both context and type of risk; governance risk has the largest impact on short-term volatility; environmental risk lines up with long-term structural instability; and social risk operates by sector/industry and reputation.

ESG should be viewed as not only direct predictors but rather as risk-structuring agents which create vulnerability to financial performance for firms and industries.

VIII. IMPLICATIONS

A. Academic Implications

The contribution of this research consists of moving from composite score approaches within the ESG-finance literature toward interpreting ris within dimension-level modelling at a higher level of granularity. In doing so, the research presents empirical evidence supporting the view that ESG matters to financial markets primarily through structural channels as opposed to linear causality. The use of explainable machine learning provides new theoretical development opportunities through the conversion of black box predictions back into observable economic relationships.

From a methodological perspective, the research demonstrates that explainability methods (e.g., SHAP-based decompositions) serve as tools for developing theory, rather than simple model diagnostics.

B. Practical Implications

With regards to investment/risk management practices, the research indicates that ESG data should not be used as stand alone screening criteria, but rather ESG data should be used within a multi-factor risk context in order to provide risk contextualization, with ESG signals being of greater value and meaning when viewed in relation to the overall/totality of sectoral structure and what effects occur by way of the interaction between factors.

For regulators/policy makers, the research indicates that governance and environmental risk channels are structural/systemic in nature, providing further evidence to support the incorporation of ESG into prudential regulation and stress testing.

IX. CONCLUSION

In this research, we use a machine learning tool to interpret financial risks from the perspective of ESG. Specifically, we use decomposed ESG dimensions to model sectors while leveraging explainable AI to analyze how ESG is driving financial risks.

We do not measure ESG's impact through prediction scores but rather through how these ESG dimensions impact financial risk structurally.

Our analysis found that each of the ESG dimensions uniquely contributes to financial risk differently than suggested by aggregate scores—governance risk has an immediate impact on volatility; environmental risk has a long-term impact on structural instability; and social risk will be measured differently from sector to sector.

Thus, combining predictive modeling and explainability has transformed the analysis of ESG from a correlation-driven prediction to a mechanism-driven understanding of financial risk, with explainable AI serving as an integral tool to support research and governance related to financial risk.

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